MOI HIGH-KABARAK 2025 PREDICTION

Kenya Certificate of Secondary Education

KCSE PREDICTION SERIES 1

447/2 **Power Mechanics** Paper 2

Time: 2½ Hours

NAME………………………………………………………………………………………

SCHOOL………………………………………………… SIGN…………………………

INDEX NO……………………………………………… ADM NO………………………

Kenya Certificate of Secondary Education.

447/1

POWER MECHANICS

PAPER 2

TIME: 2½ HOURS

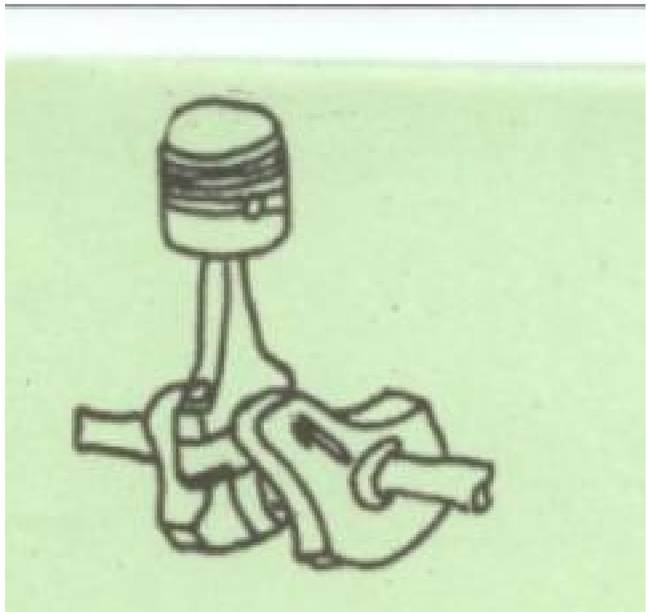
INSTRUCTIONS TO CANDIDATES

1. *There are ten stations in this examination.*
2. *Candidates are allowed 15 minutes at each station .*
3. *At each station ,candidates are not allowed to either review the previous station work or read instructions for other stations.*
4. *Write your name and index number in the space provided at the top of this page .*
5. *Attempt all the exercise in each station.*
6. *All dimensions in millimeters.*

STATION 1

Instructions:

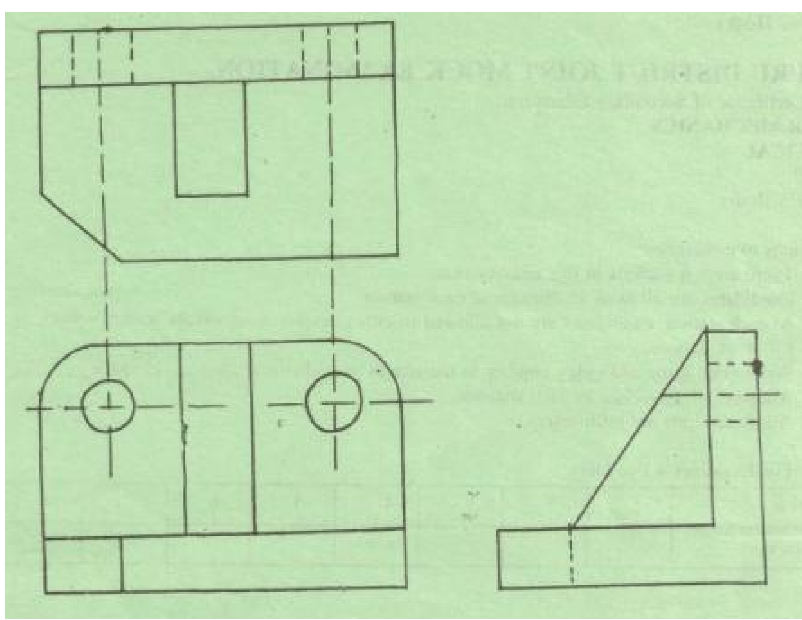
The figure below shows a piston assembly of a single cylinder engine. In the space below, sketch an exploded view of the assembly and label all the parts. **(10 marks)**



STATION 2

Instructions

The figure below shows a shaped block drawn in third angle orthographic projection. In the space provided below, draw the oblique view of the block in good proportion.(10 marks)



STATION 3

Instructions:

From the manual provided, look for and record all the information required to complete the following table for models HONDA G35 (APPENDIX A) and CLINTON ENGINE (VS 2100)(10 marks)

HONDA G35 (Standard)

1.Exhaust valve stem O/D ………………………

2. Valve spring free length ……………………….

3. spark plug gap. …………………………………

4. Piston O/D at skirt ………………………..

5. Cam height …………………………………….

CLINTON VS 2100

1. Carburetor float setting ……………….max
2. Ring end gap in cylinder ……………….max
3. Spark plug gap ……………………min
4. Piston O.D. at skirt ……………..min
5. Point setting …………………

STATION 4

Instructions

Make a good proportion sketch of a two-leading shoe design of a drum brake on the space provided below and also label it .

STATION 5

Instructions:

Identify the tools labeled A to J and state the use of each. (10 marks)

TOOL	NAME	USE
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		

STATION 6

Instructions

Using the conventional ignition system components and the wires provided. Connect a complete ignition circuit as found in a vehicle**.(10 marks)**

Let the examiner inspect your work and disconnect the circuit.

STATION 7

Instructions

The single cylinder engine provided has a clearance volume of 48cm3.

i) Measure the bore……..mm

ii) Measure the stroke……mm

Calculate iii) the cubic capacity of the engine

iv) The compression ratio of the engine

v) The amount of fuel the engine consumes in 10 minutes if its air fuel ratio is 15:1 and the engine is running at 1200 has a volumetric R.P.M and efficiency of 80%.

STATION 8

Instructions: Identify the components numbered 1 to 10 and complete the table given below. (10 marks)

NO.	Name of component	one use
1		
2		
3		
4		
5		
6		

7
8
9
10

STATION 9

Instructions

The engine provided has faults in its basic system indicated below. Identify the faults and state the effect of each on the performance of the engine. (10 marks)

SYSTEM	FAULTS	EFFECTS
a) Fuel		
b) Ignition		
c) Lubrication		
d) Cooling		
e) Exhaust		

STATION 10

Instructions

On the gas welding equipment provided: a) identify the parts labelled A to E

A
B
C
D
E

b) Light the equipment and set the torch to a carburizing flame.
(NOTE: The regulator pressures are already set).
Let the examiner check your work. **(10 marks)**